



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

SUTD 50.040: Natural Language Processing Custom Project

001

002

Multi-Layered Ideological Stance and Bias Analysis in News Articles Through Fine-Tuned Large Language Model

003

004

Elliot Phua

Computer Science and Design
1006967@mymail.sutd.edu.sg

Palinya Sengdalavong

Computer Science and Design
1006926@mymail.sutd.edu.sg

Ernest Tan

Computer Science and Design
1006883@mymail.sutd.edu.sg

005

Tejaswini Venketroyalu

Computer Science and Design
1007014@mymail.sutd.edu.sg

Amos Tan

Design and Artificial Intelligence
1006640@mymail.sutd.edu.sg

006

007
008
009
010
011
012
013
014
015
016
017
018
019
020
021
022
023
024
025
026
027
028
029
030
031
032
033
034
035

036
037
038
039
040
041
042
043
044
045
046
047
048
049
050
051
052
053

054
055
056
057
058

Abstract

Digital media polarization has amplified confirmation bias and eroded trust by subjecting news consumers to algorithmically reinforced ideological echo chambers (DeVos, 2021). Current monitoring systems, such as Ground News, fail to intervene effectively because they rely on static, source-level bias ratings, neglecting the real-time, nuanced linguistic framing in individual articles (Ground News, 2025b). This project, **PoliSense**, addresses this research gap by serving as an *AI-powered Multi-layer Political News Analyzer*. We propose fine-tuning a powerful and commonly used Transformer-based language models, RoBERTa (Liu et al., 2019), utilizing Adam optimizer from Kingma and Ba, 2015, to perform: **Ideological Stance Classification** (Left/Right/Neutral) and **Bias Classification** tasks (Non-biased/Biased/No agreement). We explored several training techniques including general fine-tuning, sequential transfer learning and multi-task learning (MTL), in which the performance of each model is compared and thoroughly analyzed. Our central contribution is an end-to-end solution that allows reader to check for Multi-layer Ideological Disentanglement and framing bias through interactive and comprehensive interface.

1 Introduction

1.1 Project Overview and Motivation

The rapid consumption of news through online media, particularly in the sociopolitical domain, has amplified the impact of ideological bias. Users are increasingly subjected to content within algorithmic echo chambers, making it difficult to gauge the objective position or sentiment expressed towards political subjects (DeVos, 2021).

This project proposes the development of **PoliSense: AI-powered Multi-layer Political News Analyzer**, an end-to-end NLP application designed to identify and analyze the inherent bias and partisan ideology using multiple fine-tuned Transformer-based models. It also provides a clear, quantifiable interpretation of the linguistic indication of bias, addressing a significant gap in many existing platforms.

1.2 Problem Statement

With the lack of platforms that can provide linguistically transparent and granular bias and political leaning classification of individual news articles, we aim to determine

"How might we develop a real-time platform that can classify - (1) ideological stance and (2) framing bias - in news articles reliably, and provide readers with valuable insights into each article's bias, such that readers can evaluate the article's information objectively."

2 Related Work

2.1 Traditional Media Bias Detection and Static Ratings

The mentioned challenge has been widely explored and addressed in multiple ways, yet there are still gaps to be filled. Fundamentally, to overcome an implicit bias, one needs to be aware of it (Lee, 2017). Many systems estimate bias at the outlet or source level using aggregated human ratings and expert judgment of thousands of news articles world-wide, so they can only be shown as static bias charts or scores for users (AllSides, 2025). One such prominent provider, Ground News, aggregates more than 30,000 articles from 50,000 news outlets daily. It then manually rates articles on political leaning and reliability based on the third-party independent organizations dedicated to monitoring and rating news publishers (Ground News, 2025b). This information is then visualized in the signature Bias Bar that instantly reveals the political leaning of a story through percentage scoring (Ground News, 2025a).

Crucially, however, the very mechanism that delivers this insight also imposes a limitation. Ground News, 2025a's displayed percentages are calculated strictly by counting the number of bias-rated sources reporting on a story; this surface-level methodology fails to provide readers with the necessary granular details or an in-depth analysis of why a particular article received its specific sorting or score (Wang et al., 2025). Recent surveys highlight several limitations of the source-level bias classification approach: ratings are inherently non-static because outlets can shift their editorial line over time, static outlet labels ignore intra-outlet variation across individual stories, and coarse bias categories capture neither sentence-level framing nor intensity (Nakov et al., 2024).

2.2 Transformer-based Stance and Sentiment Analysis

The emergence of large language models (LLMs) has revolutionized the task of classifying text and stance detection by offering new capabilities to

059
060
061
062
063
064

065
066
067

068
069
070
071
072
073
074
075
076
077
078
079
080
081
082
083
084
085
086
087
088
089
090
091
092
093
094
095
096
097
098
099
100
101
102

103
104
105
106
107

understand complex opinions, generalization between domains, and multi-modal stance analysis (Pangtey et al., 2025). Transformer architectures such as BERT, RoBERTa, and DeBERTa have now become the de-facto standard for political stance, sentiment, and ideology prediction from text, consistently outperforming earlier feature-based and RNN models (Wessel, 2022). Work on political orientation in news demonstrates that fine-tuned RoBERTa models achieve strong performance for multilingual ideological classification, propaganda detection, and related tasks, often surpassing much larger generative LLMs on supervised classification benchmarks (Palmqvist et al., 2024), underscoring their immediate viability and established superiority for tackling specific classification challenges like those investigated in this study.

3 Methodology

3.1 Tasks Definition.

PoliSense will be designed to perform two core Natural Language Processing (NLP) tasks: (1) **Ideological Stance Classification** and (2) **Bias Classification**.

3.1.1 Ideological Stance Classification

The first NLP task focuses on determining the political orientation of a given article or sentence by classifying it into one of three categories: Left, Right, or Center (3-Class nominal classification). All article and sentence-level models experimented are trained on the same base RoBERTa architecture, using *Traditional Fine-Tuning* technique. The general ideological stance of the entire article is the primary output of the application, while the ones from sentence-level variant of the model will be used to support the bias classification task that will be discussed subsequently in 3.1.2.

3.1.2 Bias Classification

The second NLP task focuses on detecting the presence of linguistic or, also know as, framing bias within a sentence, regardless of its ideological stance. Sentences will be classified into one of three distinct categories: Biased, Non-biased, or No Agreement (3-Class nominal classification). One *sentence-level bias* model will be trained using *Sequential Fine-Tuning*, the transfer learning technique, on the previously mentioned sentence-level ideological leaning model (3.1.1) to output the bias of the sentence and its confidence level as shown in figure 1. PoliSense then highlights the top five

sentences most likely to contain bias using the confidence scores, enabling users to closely examine how specific phrasing or framing may shape their interpretation.



Figure 1: PoliSense Models Training Pipeline.

3.2 Data Collection

To allow PoliSense’s models to execute the **two** tasks accurately and efficiently, four pre-labeled datasets are used to train our models to recognize political news framing:

- (i) **Bias Annotations by Experts (BABE):** BABE contains 2000 sentence-level annotated data points for bias categories created by expert annotators across multiple political articles (Spinde et al., 2021). This dataset includes SG-1 and SG-2 subsets which labels linguistic biases at a high granularity.
- (ii) **We Can Detect Your Bias:** Predicting the Political Ideology of News Articles (Baly et al., 2020). This dataset contains 37500 article-level annotated data points for ideological leanings based on political news sources.
- (iii) **Annotating and Analyzing Biased Sentences in News Articles using Crowdsourcing:** We introduced sentence-level bias judgment from performing crowd-sourcing (Lim et al., 2020). This dataset includes 800 sentence-level annotations of bias, where labels are derived from public perceptions instead of expert-defined standards.

- (iv) **Media Bias Identification Corpus (MBIC):** MBIC provided us with 1700 sentence-level annotations for political leaning, opinion level, and bias spans across a broad range of news sources (Kaggle, 2024). MBIC explicitly includes labels for political stance at both sentence and article levels, while also covering mainstream and fringe outlets.

3.3 Data Processing

Filtering: Given that the datasets were pre-collated and contained standardized annotations, no additional filtering was required prior to feature selection.

Feature Selection: We performed dataset-specific feature selection to retain only the text and label fields relevant for our NLP tasks. The extracted features for each dataset are as follows:

- **Dataset (i) and (iv):** We retained the text, type, and label_bias columns. These represent the sentence content, the political leaning (Left, Center, Right), and the bias classification (Biased, Unbiased, No Agreement).
- **Dataset (ii):** We selected the content and bias_text fields, corresponding to the full article text and its ideological label (left, center, right).
- **Dataset (iii):** We extracted the document title (doctitle) and the sentence fields (s1–s19), along with their associated bias labels (t and 1–19). The column t denotes the title’s bias rating, while columns 1–19 correspond to sentence-level ratings. All bias ratings use a 4-point scale where 1 = Unbiased and 4 = Very Biased.

Aggregation: For Dataset (iii), as each article was annotated by five distinct crowd-sourced participants, we aggregated the sentence-level scores. The final bias score for each sentence was derived by calculating the *mean* of the five labels and rounding the result to the nearest integer.

Normalization and Formatting: To ensure interoperability across datasets, we normalized the labels and standardized the data format:

1. **Label Standardization:** For Dataset (iii), the aggregated scores were binarized: a score of 1 was mapped to "Unbiased," while scores of 2, 3, and 4 were mapped to "Biased." Subsequently, we aligned Datasets (i), (iii), and (iv)

to a unified numerical schema: 0 (Unbiased), 1 (Biased), and 2 (No Agreement).

2. **Data Structure:** We reorganized Datasets (i), (iii), and (iv) into a consistent JSON format:

```
{
  "text": "...",
  "label": "0"
}
```

For Dataset (ii), headers were renamed to facilitate ideological prediction, resulting in the following structure:

```
{
  "text": "...",
  "label": "Center"
}
```

3.4 Implementation of Classification Models

To address the multi-granular NLP requirements of PoliSense, we trained a suite of eight transformer-based models to evaluate which methods would yield the best results. These models cover different combinations of task type (ideological stance vs. bias), scope (article-level vs. sentence-level), and weight initialization strategy. All models use the pre-trained RoBERTa architecture as the backbone.

1. Baseline Stance Models (M_1 – M_4): We developed four stance classifiers tailored to different input lengths and dataset sources. M_1 processes full articles, which includes the headline and body paragraphs, using Dataset (ii). Models M_2 , M_3 , and M_4 perform sentence-level stance classification using Dataset (iii), a concatenation of Datasets (ii) and (iii), and Dataset (iv), respectively. All four models were directly fine-tuned on pre-trained base RoBERTa.

2. Sequential Transfer Learning Models (M_5 , M_8): To leverage domain-specific representations, we implemented two sequential fine-tuning pipelines:

- M_5 (**Sentence Stance**): Initialized from the article-level stance classifier M_1 , then fine-tuned on Dataset (iii).
- M_8 (**Sentence Bias**): Initialized from the sentence-level stance classifier M_4 , then fine-tuned on the bias-specific Dataset (i).

3. Bias and Multi-Task Models (M_6 , M_7): We developed a multi-task architecture, M_6 , capable of

jointly predicting stance and bias. M_6 was trained on a combined corpus of Datasets (i) and (iv), with weight initialization from M_1 . M_7 is a dedicated sentence-level bias classifier trained on Dataset (i).

Final System Pipeline: After evaluating all model variants, the final PoliSense pipeline was instantiated using a composite configuration. We deployed M_1 for article-level political stance classification and M_7 for sentence-level bias detection, leveraging its transfer learning path from M_4 to maximize feature retention and improve generalization.

4 Results and Performance Evaluation

Model	Level of Fine-tuning	Dataset(s)	Accuracy (%)	F1
M_1	Article-Leaning	(ii)	90.6	0.949
M_2	Sentence-Leaning	(iii)	61.8	0.600
M_3	Sentence-Leaning	(ii) + (iii)	60.6	0.609
M_4	Sentence-Leaning	(iv)	90.6	0.923
M_5	Sentence-Leaning(pre-trained on article-leaning)	(ii) -> (iii)	90.0	0.925
M_6	Multitask Learning(pre-trained on article-leaning)	(ii) -> (i) (iv)	80.1 (bias), 92.1 (leaning)	0.545 (bias), 0.697 (leaning)
M_7	Sentence-Bias	(i)	75.8	0.514
M_8	Sentence-Bias(pre-trained on sentence-leaning)	(iv) -> (i)	77.3	0.530

Table 1: Accuracy & F1 scores of the models trained and evaluated.

Task	Dataset(s)	Accuracy (%)	F1
Article-Leaning	(ii)	45.1	0.211
Sentence-Leaning	(iv)	43.4	0.204
Sentence-Bias	(i)	48.7	0.218

Table 2: RoBERTa’s performance without fine-tuning.

4.1 Article-Level Ideology Classification

We give an overview of our evaluation results for both article-level and sentence-level ideological

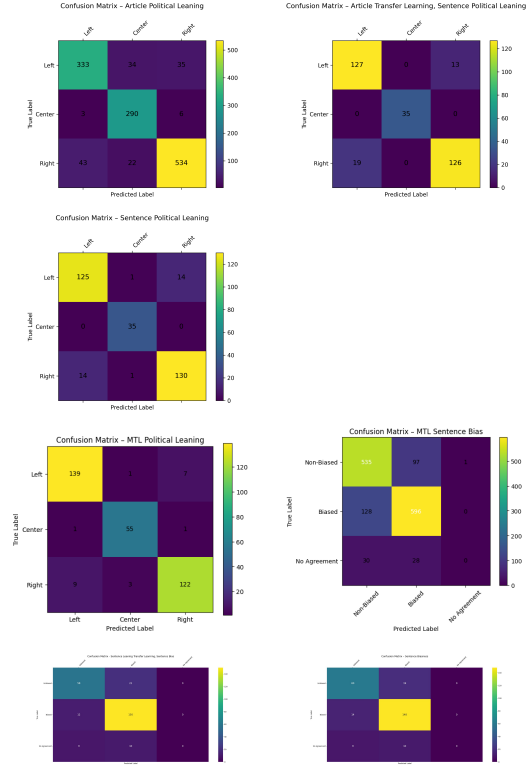


Table 3: Confusion matrices of the evaluation of fine-tuned models.

stance prediction. As seen in table 1, the article-level leaning classifier has an accuracy of approximately 90.6% which shows that the model is able to correctly distinguish between Left, Center, and Right leaning articles in majority of cases.

Moreover, the fine-tuned sentence-leaning models achieved a comparable accuracy of 90.6% which shows that ideological signals are often detectable at sentence-level. When we initialized the sentence-leaning classifier with our scoring/weight mechanism from the article-leaning model, we noticed that there had been an improvement in accuracy to 92.1%, confirming that the coarse article supervision benefits fine-grained ideological prediction via transfer learning.

4.2 Bias Classification

The bias classification was evaluated using three classes: “non-biased”, “biased”, and “non-agreement”. The standalone model achieves a bias-classification of around 80%, although macro-F1 is lower at around 0.55% which is probably due to class imbalance and the greater linguistic availability that is associated with subjective expressions. This gap between accuracy and F1 shows that it remains a challenge to correctly identify strong

biased instances and distinguish neutral content.

4.3 Ablation Study and Baseline Comparison

To assess the effect of multi-task learning, we performed an ablation whereby we trained ideological stance and bias classification both independently and jointly. Training for sentence-level ideological stance on Datasets (ii) and (iii), we found that the results were not as high as expected, obtaining an accuracy of about 60%. When both tasks are trained together, the model preserves high ideological leaning accuracy of 92% while maintaining competitive bias classification performance of 80%.

Compared with single-task baselines, the multi-task setup yields:

- Improved sentence-level ideological accuracy (2% increase)
- Stable bias performances
- Reduced total training time due to shared feature representations

These results show that sentence-level ideological stance and bias are mutually informative, and that multi-task learning enables the model to generalize across related rhetorical phenomena without sacrificing task-specific accuracy.

Despite the promising results of the MTL model, we decided to use two separate models: (1) the **Article-Leaning model** to tackle the political stance classification task, and (2) the **Sentence-Bias (Transfer Learning From Sentence-Leaning) model** for the bias classification task. This was because the MTL model required greater engineering efforts in implementation.

5 PoliSense User Workflow and Features

The final measure of the PoliSense system lies not just in its predictive accuracy, but in its ability to translate complex, low-level model inference into high-value, actionable insights for the user. This section details the end-to-end process, from raw article ingestion to the interactive visualization presented in the Analysis Results Dashboard emphasizing how the architectural choices directly facilitate transparency and intervention.

5.1 End-to-End Workflow

The PoliSense workflow is a streamlined process from raw text ingestion to diagnostic output. It begins with the article input being extracted, tokenized and segmented for the Transformer Backbone. The fine-tuned, Adam-optimized models simultaneously execute Political Stance Classification and Bias Classification through its specialized output heads. The confidence score and ranking mechanism are then generated to aggregate sentence-level results into the overall Article Political Leaning. PoliSense also displays the confidence score (out of 10) of sentences classified as biased in the article. Finally, the distilled metrics and AI Rhetorical Analysis are presented on the Analysis Results Dashboard, ensuring near real-time, deep linguistic transparency.

5.2 Confidence in Bias Identification

PoliSense explicitly counters the "black box" critique (Blouin, 2023) by quantifying its analytical certainty. The AI Confidence Score (e.g. 85%) is a meta-cognitive metric derived from the underlying model probabilities. This score then provides transparency, acting as a direct measure of prediction reliability. High confidence validates the platform's judgment, while low confidence flags ambiguity due to mixed signals or complicated framing, compelling the user toward a necessary, cautious interpretation. This feature makes use of a probabilistic, evidence-based approach to media scrutiny.

5.3 Real-World Demonstration

The Analysis Results Dashboard serves as the definitive demonstration of the system's core value. It visually presents the outcome of the Political Stance Classification via the Article Political Tier. Crucially, the system isolates Key Influential Sentences accompanied by their Confidence Scores (e.g. 9/10), also known as Impact Scores, which are the direct outputs of the Attribution-based Bias Classification. This functionality offers precise and tangible evidence, proving that PoliSense successfully facilitates Active Linguistic Intervention by moving analysis from abstract labels to concrete textual proof.

6 Discussion

Our models show that fine-grained ideological stance and bias classification can be learned. However, there exist several limitations that constrain

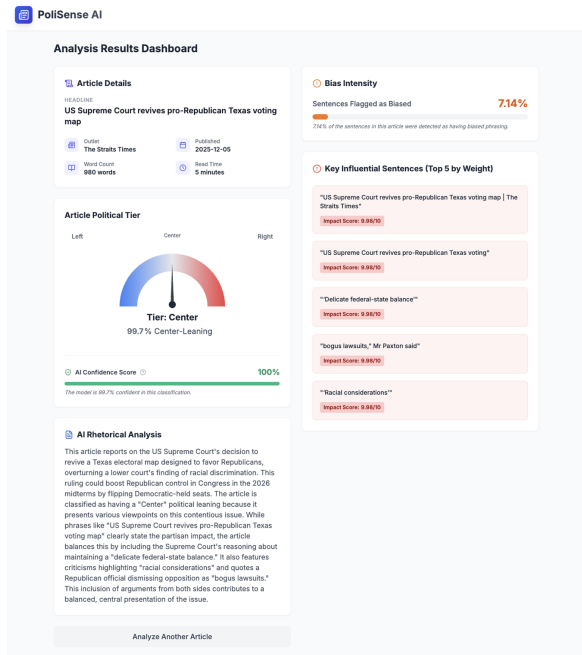


Figure 2: PoliSense Dashboard.

the generalizability and interpretability of the results.

6.1 Challenges and Areas of Improvement

6.1.1 Data Limitations and Label Quality

One major limitation includes the relatively small training samples for bias classification, particularly within the BABE subsets, where there is a disproportionate distribution between the different bias classes (Spinde et al., 2021). This translated to class imbalance, which was reflected in the lower macro-F1 scores despite competitive accuracy. Moreover, the concept of “bias” lacks a universal agreement. Previous studies display the variation across annotators with different ideological backgrounds or levels of expertise. For example, the expert-labeled datasets (Spinde et al., 2021) differed significantly from crowd-sourced annotation (Lim et al., 2020), which ultimately raises concerns about annotation reliability and perceived bias subjectivity.

6.1.2 Methodological and Model Architecture Constraints

Our experiments revealed that *Transfer Learning* improves sentence-level leaning classification, however, this is less so for bias classification. This may indicate that ideological stance is more linguistically salient and consistently annotated than bias classification. Moreover, our reliance on sentence-level features means that our model does not fully

leverage richer discourse-level signals that span cross-sentence context.

6.1.3 System Scalability and Real-World Application

Another noticeable limitation includes the problem that the datasets used are predominantly US-centric, focusing only on American media outlets and political discourse. This could result in our model overfitting to US ideological patterns, thus not generalizing well to other geopolitical contexts. This hence constrains the applicability of PoliSense outside Western media ecosystems, highlighting the need for multi-country datasets with culturally diverse annotation schemes.

7 Conclusion and Future Work

In conclusion, our project introduced PoliSense, an end-to-end AI system for multi-layer political news analysis using transformer-based sequential fine-tuning. Our empirical results show a strong performance for ideological classification at both article and sentence level, proving that bias classification can be jointly learned without degrading stance classification performance.

However, due to the limitations explained in section 6 of this report, there needs to be future work done to prioritize expanding training datasets, especially for bias classification. This can be done by integrating larger cross-annotated corpora or collecting new samples through controlled annotation protocols. Moreover, by incorporating annotator metadata — such as political affiliation or various demographic backgrounds, it can help disentangle annotator-specific variation and improve annotation reliability (Spinde et al., 2021). In future data pre-processing, bias classes such as “No agreement” can be filtered as it suggests that these data do not lie within the biased and non-biased spectrum, but rather that such data are outliers where human annotators are unsure of its classification. Finally, later versions of PoliSense could extend beyond US-centric sources by incorporating multilingual and cross-national datasets which enable a wider applicability and increasingly robust detection of ideological framing in global media. Future experiments can also consider the effects of the length of articles in influencing the results of the models.

References

- AllSides. 2025. [How allsides rates media bias](#). Accessed: 2025-12-10.
- Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. 2020. [We can detect your bias: Predicting the political ideology of news articles](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4982–4991, Online. Association for Computational Linguistics.
- Lou Blouin. 2023. [Ai’s mysterious ‘black box’ problem, explained](#).
- Marielle DeVos. 2021. [The echo chamber effect: Social media’s role in political bias](#). *YIP Institute*.
- Ground News. 2025a. [Bias bar](#).
- Ground News. 2025b. [Media bias](#).
- Kaggle. 2024. [Mbic - a media bias annotation dataset](#).
- Diederik P. Kingma and Jimmy Ba. 2015. [Adam: A method for stochastic optimization](#). *arXiv preprint arXiv:1412.6980*. Published at ICLR 2015.
- Cynthia Lee. 2017. [Awareness as a first step toward overcoming implicit bias](#). *SSRN Electronic Journal*.
- Sora Lim, Adam Jatowt, Michael Färber, and Masatoshi Yoshikawa. 2020. [Annotating and analyzing biased sentences in news articles using crowdsourcing](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 1478–1484, Marseille, France. European Language Resources Association.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. [Roberta: A robustly optimized bert pretraining approach](#). *arXiv preprint arXiv:1907.11692*.
- Preslav Nakov, Jisun An, Haewoon Kwak, Muhammad Arslan Manzoor, Zain Muhammad Mujahid, and Husrev Taha Sencar. 2024. [A survey on predicting the factuality and the bias of news media](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 15947–15962, Bangkok, Thailand. Association for Computational Linguistics.
- Oscar Palmqvist, Johan Jiremalm, and Pablo Picazo-Sanchez. 2024. [Ideology and power identification in parliamentary debates](#). In *Working Notes of CLEF 2024 – Conference and Labs of the Evaluation Forum*, volume 3740 of *CEUR Workshop Proceedings*, Grenoble, France. CEUR-WS.org.
- Lata Pangtey, Anukriti Bhatnagar, Shubhi Bansal, Shahid Shafi Dar, and Nagendra Kumar. 2025. [Large language models meet stance detection: A survey of tasks, methods, applications, challenges and future directions](#). *arXiv preprint arXiv:2505.08464*.
- Timo Spinde, Lada Rudnitckaia, Kanishka Sinha, Felix Hamborg, , Bela Gipp, and Karsten Donnay. 2021. [Mbic – a media bias annotation dataset including annotator characteristics](#). In *Proceedings of the iConference 2021*.
- Jenny S. Wang, Samar Haider, Amir Tohidi, Anushkaa Gupta, Yuxuan Zhang, Chris Callison-Burch, David Rothschild, and Duncan J. Watts. 2025. [Media bias detector: Designing and implementing a tool for real-time selection and framing bias analysis in news coverage](#). In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI’25)*.
- Martin Wessel. 2022. [Improving media bias detection with state-of-the-art transformers](#). Master’s thesis, University of Konstanz, Konstanz, Germany. Master’s thesis.